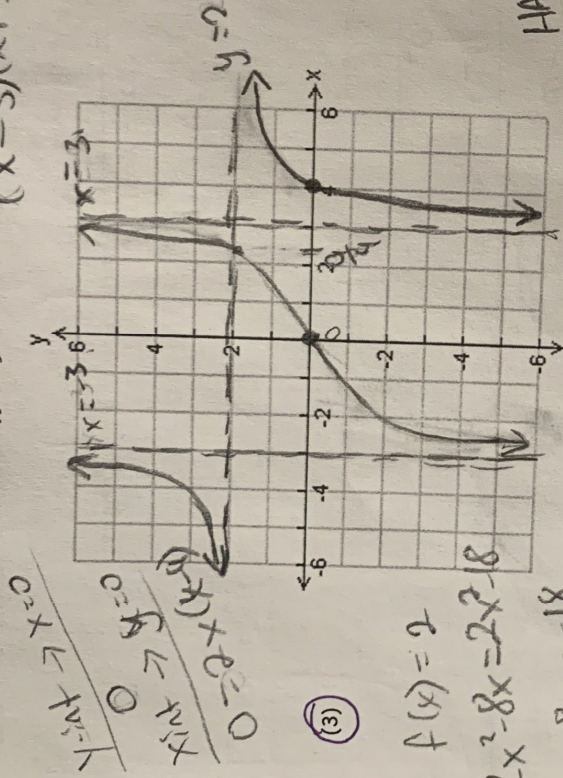


Application

1. Sketch a graph to represent each function. Be clear and label all asymptotes and intercepts.

a) $f(x) = \frac{2x^2 - 8x}{x^2 - 9}$



$f(x) = 2$

$2x^2 - 8x = 2x^2 - 8$

$-8x = -18$
 $x = \frac{9}{4}$

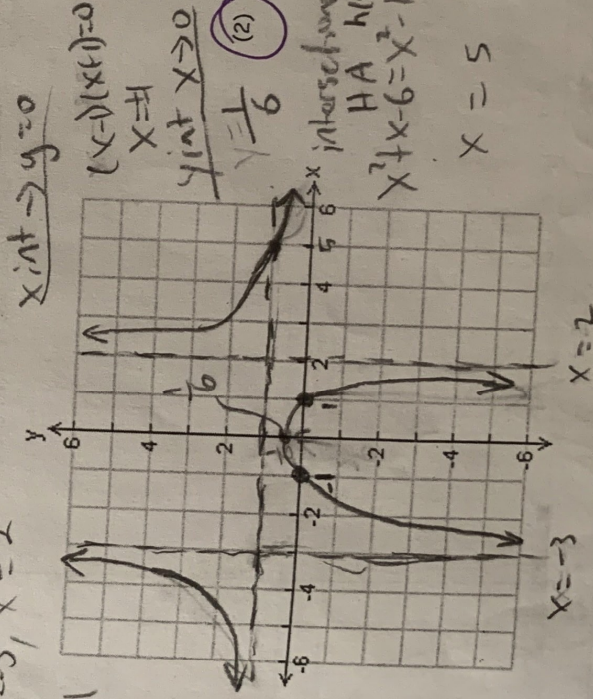
VA $x = 3, -3$

HA $y = 2$

c) $h(x) = \frac{x^2 - 1}{x^2 + x - 6}$

VA $x = -3, x = 2$

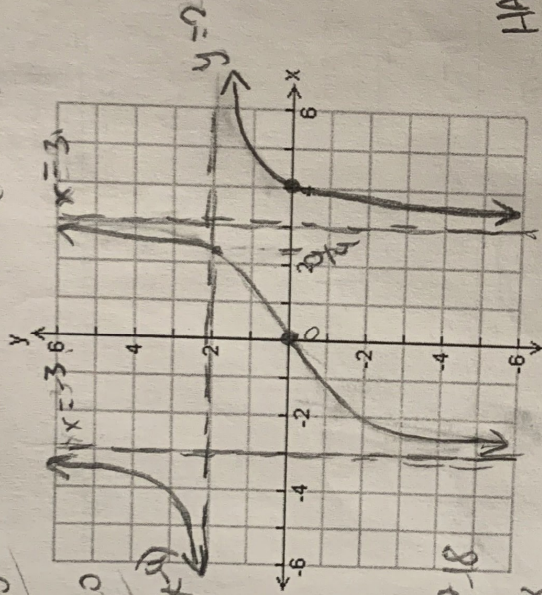
HA $y = 1$



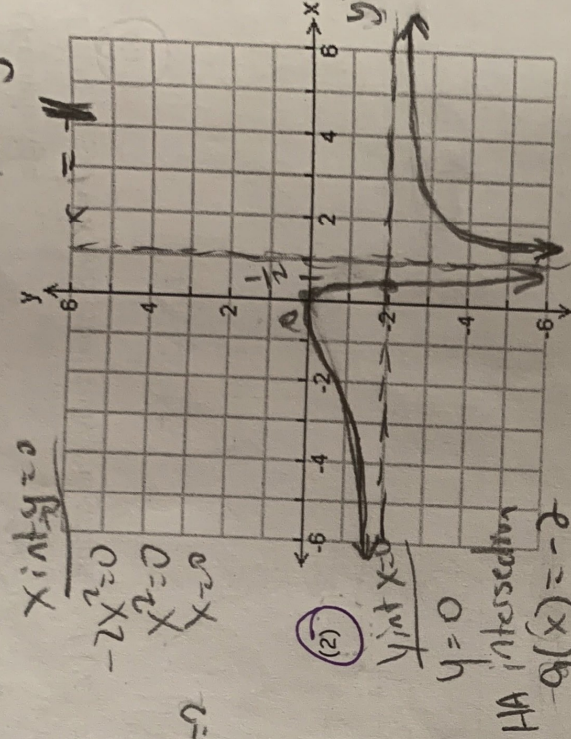
End Behaviour:	$x \rightarrow \infty$	$f(x) \rightarrow 1^-$
	$x \rightarrow -\infty$	$f(x) \rightarrow 1^+$

(1)

b) $g(x) = \frac{2x(x-4)}{(x^2-3)(x+3)}$



b) $g(x) = \frac{-2x^2}{(x-1)^2}$



$-2(x^2 - 2x + 1) = -2x^2$

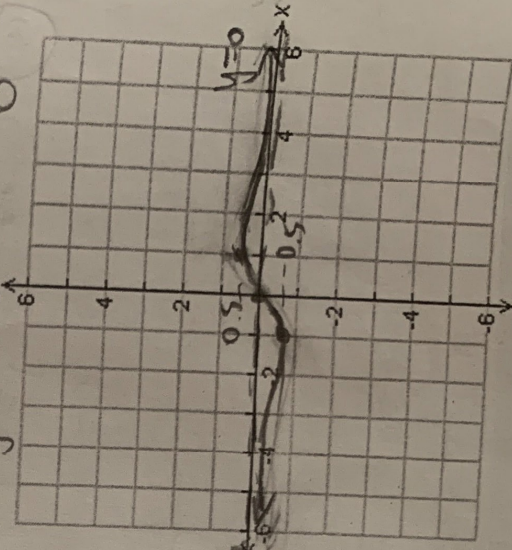
$4x - 2 = 0$
 $x = \frac{1}{2}$

$x \rightarrow y = 0$

$y \rightarrow x = 0$

VA N/A
HA $y = 0$

d) $j(x) = \frac{x}{x^2 + 1}$



$j(1) = 0.5$

$j(-1) = -0.5$

Intersection with HA
 $j(x) = 0$
 $x = 0$

$(x^2 + 1) \pm (x) (2x)$
 $x^2 + 1 - 2x^2$
 $x = \pm 1$

Intersection with H.A.

$$\frac{x-1}{(2x-7)(x+3)} = 0$$

Knowledge

1. Sketch and label fully the graph of $f(x) = \frac{x-1}{2x^2-x-21}$ and complete the chart:

$$f(x) = \frac{x-1}{(2x-7)(x+3)}$$

$$V.A. at \quad x = \frac{7}{2} \text{ and } x = -3$$

$$x_{int} \rightarrow y=0$$

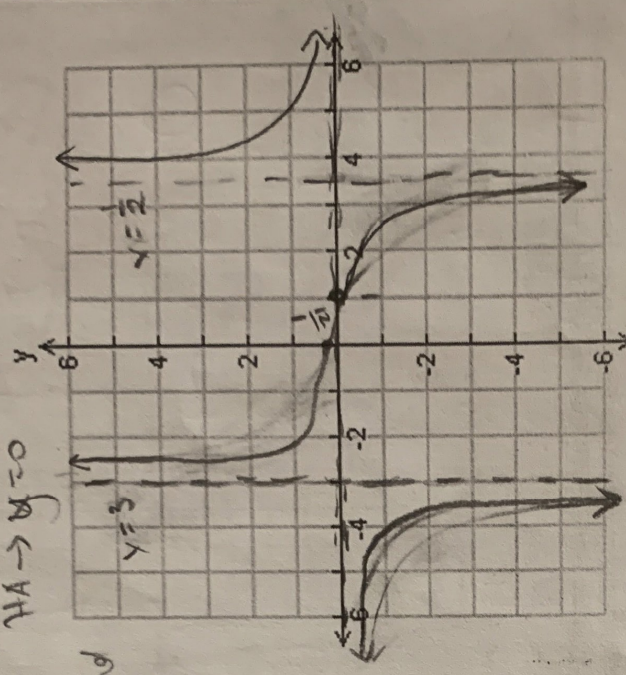
$$0 = x-1 \quad x=1$$

$$y_{int} \rightarrow x=0$$

$$\frac{-1}{-21} = \frac{1}{21}$$

End Behaviour:	$x \rightarrow \infty \quad f(x) \rightarrow 0^+$ $x \rightarrow -\infty \quad f(x) \rightarrow 0^- \quad (0.5)$
End Behaviour near vertical asymptote(s):	$x \rightarrow -3^- \quad f(x) \rightarrow -\infty$ $x \rightarrow -3^+ \quad f(x) \rightarrow \infty \quad (0.5)$ $x \rightarrow \frac{7}{2}^- \quad f(x) \rightarrow -\infty$ $x \rightarrow \frac{7}{2}^+ \quad f(x) \rightarrow \infty \quad (0.5)$

$$H.A. \rightarrow y=0$$



2. Sketch and label fully the graph $f(x) = \frac{-x}{x-1}$ and complete the chart.

$$H.A. \rightarrow y = -1$$

$$V.A. \rightarrow x = 1$$

$$x_{int} \rightarrow y=0 \quad y_{int} \rightarrow x=0$$

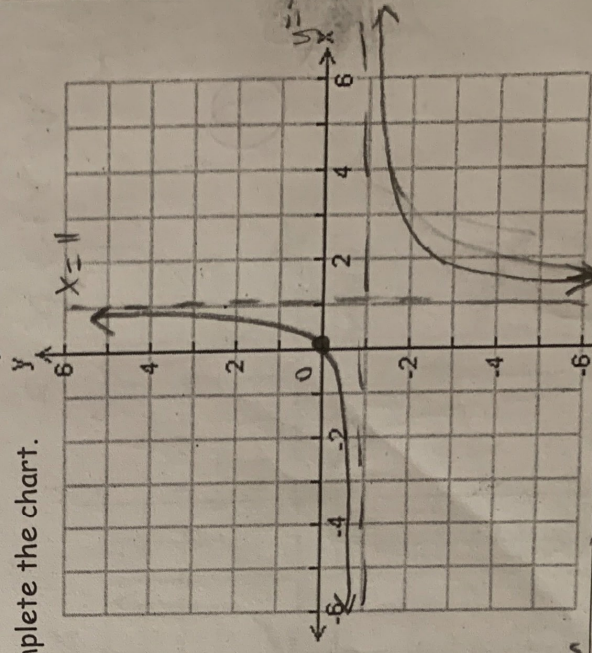
$$y=0$$

$$0 = -x$$

$$x=0$$

$$\frac{-x}{x-1} = -1 \quad \text{intersection with H.A.}$$

$$-x = -x + 1 \quad 0 = 1 \quad \text{no intersection}$$



Interval	$(-\infty, 0)$	$(0, 1)$	$(1, \infty)$
vii) Sign of $f(x)$ as $(+/-)$	-	+	+
viii) Slope $(+/-)$	+	+	+
ix) Slope (Increasing/Decreasing)	↑	↑	↓